

ROV Setup and Control

1. Connect the battery and turn on the switch
2. Connect the LAN cable to the Raspberry Pi and a host laptop which will maintain constant communication with the ROV
3. Once the Raspberry Pi has booted, connect to it via SSH or RDP with the below login details:

Computer	Username	Password
rit.local	rit	rit

4. execute the python script, available on the desktop, via the command line

```
cd Desktop
```

```
python web.py
```

 for web GUI (suggested) accessible via mobile device for remote control

```
or python main.py
```

 for local application GUI accessible via the Raspberry Pi desktop environment

5. On a mobile device, open the ROV control panel at the local URL (refer to command line) on port 5000 `192.168.1.179:5000`
6. The left and right scrollbars control the left and right motors individually. The bottom scrollbar controls the vertical thruster for resurfacing/diving. The central view is the live video feed from the ROV.

Challenges and Solutions

1. 3D design to physical structure

The initial 3D model of the ROV included joints without ridges which led to misalignment and inaccuracies between the digital and physical model during assembly. The solution to this was to shorten each PVC pipe by an amount which would match the exposed PVC length. This allowed accurate correlation between the digital and physical model.

2. Buoyancy calibration

The buoyancy of the ROV was imbalanced causing it to naturally pitch forward when in water. Additionally, the electronics container would prevent the ROV from submerging due to the greater buoyant force. The solution was iterative, first adding aluminium rods within the base pipes. The aluminium was too light hence the switch to steel rods of the same volume which allowed the ROV to naturally sink. The next was to resolve the pitch balance which was achieved by adding a steel rod at the

rear. This, however, didn't directly resolve the issue since the position along the ROV's length caused further imbalance toward the rear. The final solution to this was adjusting the position of the PVC pipe and the steel rod by moving it closer to the centre.

3. Camera access

Working with the camera connected to the Raspberry Pi was difficult as the original interfacing library was not responding and throwing connection errors. This problem occurred due to a fault in the original cable where excessive bend prevented signal transfer. This issue was resolved by using a higher quality and more flexible camera cable. The second reason causing this issue was an improper camera configuration set in the system boot config file. I was able to resolve this issue by getting the required configuration properties from the camera documentation online and adding them to the system boot config.